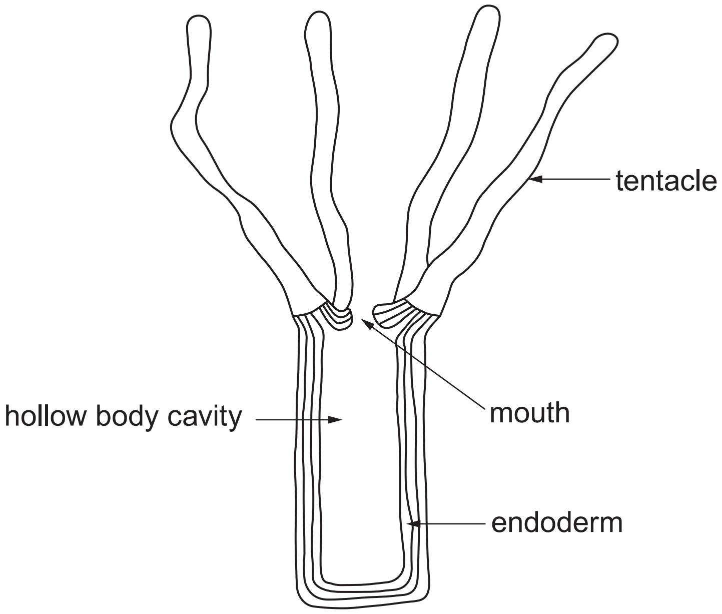
Examiner only

2. Nutrition is the term used to describe how living organisms obtain the molecules from which they build up their organic compounds. A major difference between many types of living organism is their method of nutrition.

(a) The diagram below illustrates the structure of *Hydra viridis*.



Hydra viridis is a simple, multicellular, freshwater animal that uses its tentacles to capture small organisms and transfer them through the mouth into the hollow body cavity. Gland cells in the endoderm secrete enzymes which digest the prey. The products of digestion are absorbed and indigestible remains are egested through the mouth.

(i) State the method of nutrition, exemplified by *Hydra*, where food is ingested and then digested internally. [1]

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(ii) The endodermal cells of *Hydra viridis* contain cells of the green alga *Chlorella*. This is called mutualism which is a relationship between two different species where each individual benefits from the activity of the other. Explain how both *Hydra* and *Chlorella* may benefit from this relationship. [2]

Hydra: ....

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Chlorella: ....

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(b) *Nostoc commune* and *Nitrosomonas europaea* are bacterial species that can be found in soil. Both species use simple inorganic molecules to build up their organic compounds. Chlorophyll pigments are found in the cells of *Nostoc* but not in the cells of *Nitrosomonas*.

What conclusions can be made about the methods of nutrition in these two species? [3]

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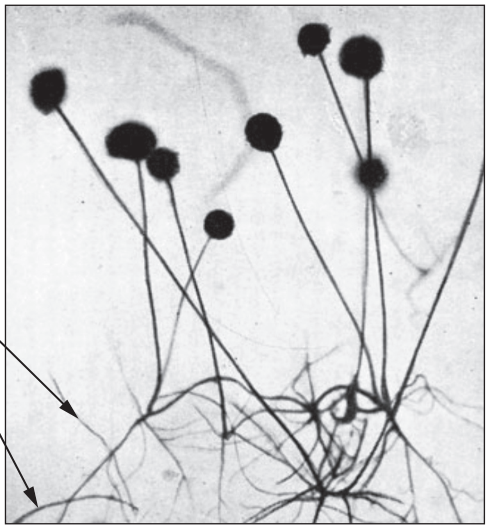
(c) The photograph illustrates the structure of a fungus belonging to the genus *Rhizopus*. All *Rhizopus* species have a similar structure.



Two species of this genus are *Rhizopus stolonifer* and *Rhizopus oryzae. R. stolonifer* is commonly found on bread surfaces and rotting fruit. *R. oryzae* can cause a rare and potentially life-threatening infection of humans called mucormycosis.

What conclusions can be made about the methods of nutrition in these two species? [4]

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